
IS1901: MICRO-CONTROLLER BASED ICT PROJECT

Project Proposal Report Level 01

AbacusGenius-1.0

Submitted by: ITM - Group 9 (Micro-Explorers)

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1.1 Introduction

The Electric Abacus enhances the traditional abacus with modern technology, including sensors, displays, and interactive learning modes. This project aims to combine the tactile experience of moving beads with real-time numerical feedback, making arithmetic learning more engaging and effective for children.

1.2 Problem in Brief

Traditional abacuses are static tools that only offer basic arithmetic capabilities. While they are beneficial for teaching math concepts, they lack interactive features that can adapt to a child's learning progress. Children may struggle to stay engaged with a non-digital version, and the absence of real-time feedback limits their ability to learn and correct mistakes effectively. There is a need for a more advanced system that combines the physical interaction of the abacus with the power of technology to provide:

- Instant feedback on arithmetic problems.
- Interactive math challenges.
- Engaging learning experiences through a modern, digital interface.

2. Aim and Features

Aim:

To create an Electric Abacus system that incorporates sensors, a micro-controller, and a display to enhance the traditional abacus, providing a fun and interactive learning environment for children to master basic arithmetic concepts.

Key Features:

1. Interactive Learning: Children will move beads on the abacus, and the system will provide real-time feedback by displaying the current value on a screen.

2. Different Modes

1. **User-Entered Problems:** Users can manually enter math problems into the app.
2. **App-Generated Problems:** Users can ask the app to generate problems.

- **Problem Types:** Users can choose whether the problems involve number identification, addition, or subtraction.
 - **Difficulty Levels:** Users can select the difficulty level of the problems.
- 3. Real-Time Calculation:** The system will calculate and display results on a screen as beads are moved, offering visual and auditory feedback.
- 4. Visual Guidance:** An intuitive interface will guide the child through each step of the calculation, showing the correct bead positions.
- 5. Learning Game Mode:** A game mode can challenge children with timed math problems, rewarding them for solving problems correctly.
- 6. Portable:** The abacus will be designed for easy portability, allowing children to practice anywhere.

3.1 Required Components and their Functions

1. Abacus Frame:

A traditional wooden or plastic frame with rods to hold the beads.

2. Beads:

Small, colorful beads that slide along the rods. These should be of different sizes for visual clarity.

3. Sensors:

Touch sensors: To evoke from the sleeping mode.

Number pad: To get input from user.

PIR sensor: To detect whether the user is in user presence and put the device into sleep mode.

4. Micro-controller

Arduino Mega NodeMCU WeMos ESP8266 : These will process the sensor inputs and control the system's logic.

5. Display:

LED Display or LCD Screen: Used to show the numeric value or results of the arithmetic operations. It may also show math problems.

6. LED:

When it's nighttime it lights up the abacus for the convenience of the child.

7. Vibration Motor:

This will provide tactile feedback to imply the user when the problem is solved incorrectly.

8. Speakers:

For sound feedback (optional), a small speaker will produce sounds when beads are moved or when the correct answer is given.

9. Battery:

A rechargeable battery or power supply to make the system portable.

10. Encoder:

To count beads on the right corner of each row separately.

11. Neodymium Magnets:

1. **Brand:** K&J Magnetics

Model: D82-N52

Dimensions: 1/2" diameter x 1/8" thick

Grade: N52

Magnetic Field: Strong enough to cover a 19cm area

integrate neodymium magnets on the right and left corners of the abacus frame.

These magnets will prevent beads from getting stuck halfway by ensuring they move smoothly to their correct positions. The magnetic power spreads to a 19cm area only, so it will not interfere with the beads on the opposite (left) corner. This design enhances the accuracy and reliability of the Smart Abacus, providing a seamless counting experience.

2. **Brand:** K&J Magnetics

Model: D21-N52

Dimensions: 1/4" diameter x 1/16" thick

Grade: N52

Magnetic Field: Strong enough to cover a 1.75cm area without exceeding it

integrate a small neodymium magnet inside the inner ring wall of each bead. This magnet will attract tiny iron balls stored in the abacus rod, completing a circuit when the bead reaches the right corner. This innovative mechanism ensures accurate counting and real-time feedback on a digital screen.

12. Tiny iron balls:

Brand: K&J Magnetics

Model: SB2 (2mm diameter steel balls)

A 2mm to 3mm diameter ensures the balls fit comfortably inside the holes without getting stuck. This size is small enough to be easily attracted by the magnets in the beads but large enough to create a reliable contact for the enclosed circuit.

13. Connectors

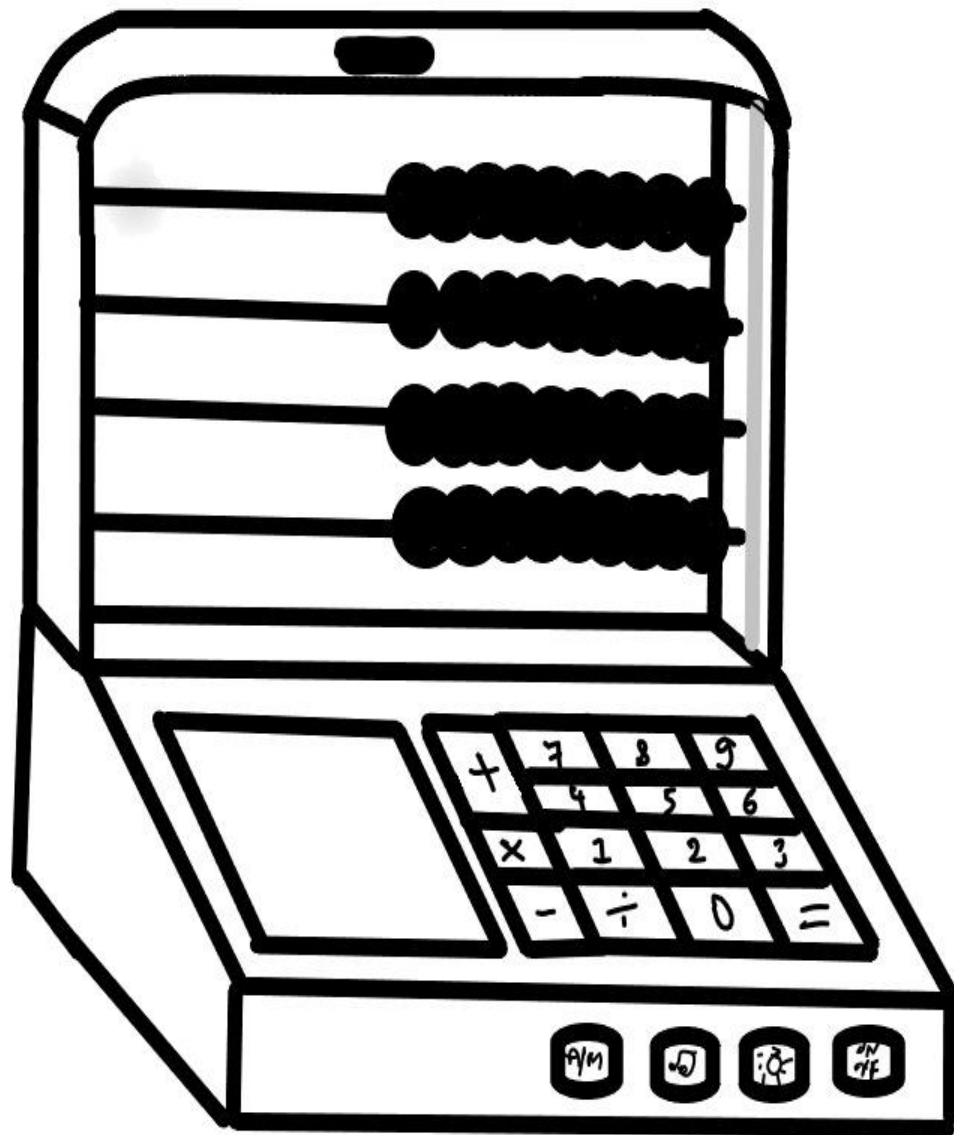
14. Mobile application:

To further enhance the Smart Abacus, we are introducing a mobile application that will significantly enrich the learning experience. The app will feature multiple difficulty levels, allowing users to progress through increasingly challenging math problems. It will keep detailed records of each user's performance, providing instant feedback and tracking their progress over time. This progress tracking will be visually represented through graphs and statistics, motivating children by showing their improvement. Additionally, the app will include math tests that adapt to the user's skill level, automatically increasing the difficulty after a certain number of correct answers.

3.2 Innovative Counting Mechanism for Smart Abacus

1. **Holes in the Rod:** Drill holes inside the abacus rod to store tiny iron balls, covered by a lid to prevent them from falling out.
2. **Magnet in Beads:** Attach a small magnet inside the inner ring wall of each bead.
3. **Magnetic Attraction:** When a bead moves to the right corner, the magnet attracts the iron ball inside the hole.
4. **Enclosed Circuit:** The attracted iron ball sticks to the lid, completing an enclosed circuit and acting as a switch.
5. **Switches for Rows:** From the number of enclosed switches, it counts the number of beads on the right corner. The encoder takes the total of enclosed circuits to find the number of beads on the right corner. Each row has its own encoder, making a total of 4 encoders. The encoder outputs the count to the digital screen, displaying it as a digital number.
6. **Encoder Output:** The encoder counts the number of beads on the right side and sends this data to the digital screen on the Smart Abacus.
7. **Passing Beads:** If a bead passes by the iron ball, it only lifts the ball up temporarily. Once the bead moves past the hole, the iron ball drops back down, breaking the circuit. This ensures that only beads in the correct position generate an output to the encoder.
8. **Linear Movement:** The inner perimeter of the bead has a small cut to prevent rolling, ensuring smooth linear movement along the rod. The rod itself is designed with a corresponding cut to match the bead's inner cut.
9. **Preventing Magnet Rolling:** Square shaped rod ensures that the magnet does not roll down with the bead, which ensures that the magnet correctly attracts the iron ball to the lid, maintaining the enclosed circuit.

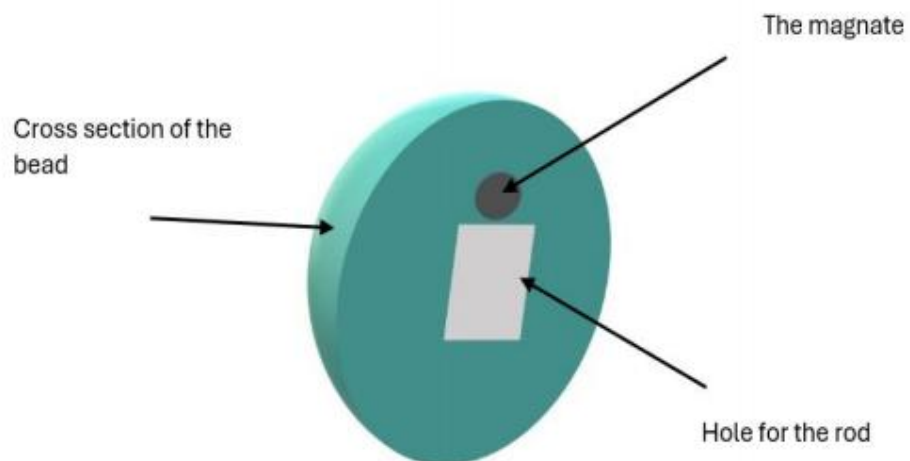
3.3 Proposed Design of the final product



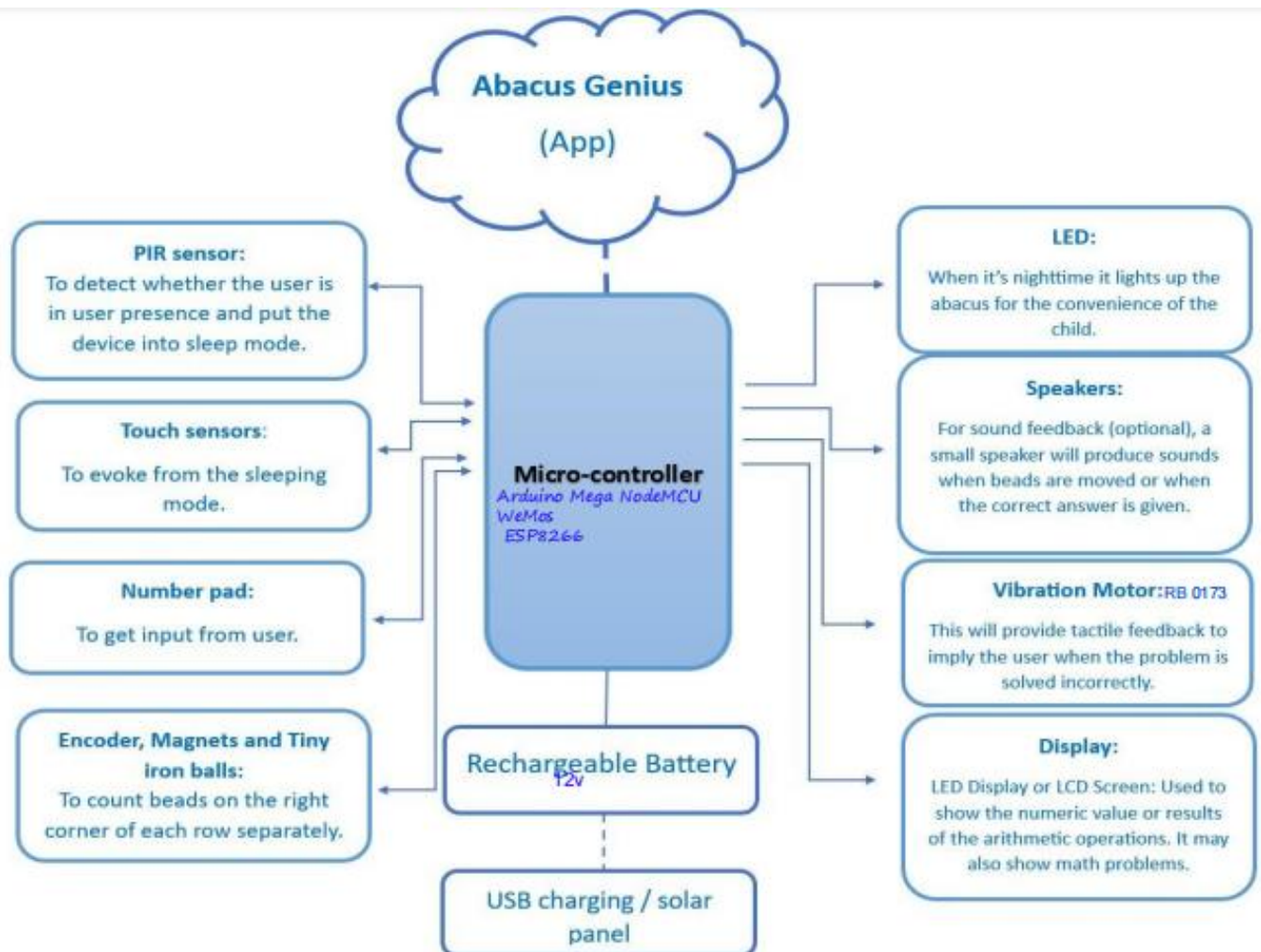
The illustration of the rod of AbacusGenius 1.0



The magnet inside the bead



3.4 Block Diagram of components



3.5 Estimated Budget for the Project

Component	Unit Price	Quantity	Price
Arduino Mega NodeMCU WeMos ESP8266	Rs.4550	1	Rs.4550.00
Encoders	Rs.70.00	4	Rs.280.00
Steel Balls	Rs.50.00	36	Rs.1800.00
Touch Sensor	Rs.320.00	1	Rs.320.00
Light Sensor	Rs.100.00	1	Rs.100.00
Neodymium Magnates	Rs.30.00	36	Rs.1080.00
N52 Disc Magnates	Rs.100.00	8	Rs.800.00
PIR Sensor	Rs.280.00	1	Rs.280.00
LED Display	Rs.400.00	1	Rs.400.00
LED Strips	Rs.350.00	1	Rs.350.00
Connectors	Rs.1500.00	As needed	Rs.1500.00
Buzzer	Rs.80.00	1	Rs.80.00
Vibration Motor	Rs.200.00	1	Rs.200.00
Speaker	Rs.50.00	1	Rs.50.00
Rechargeable Battery	Rs.520.00	1	Rs.520.00
Push Button Switches	Rs.130.00	4	Rs.520.00
Number Pad	Rs.1000.00	1	Rs.1000.00
TOTAL			Rs.14784.00

4.Conclusion

The Electric Abacus offers an innovative approach to teaching children basic arithmetic skills through interactive and engaging learning methods. By combining the traditional principles of the abacus with modern technology, the electric abacus can provide real-time feedback, challenges, and learning modes that keep children engaged while learning math. This project can be a valuable educational tool for both classroom and home use, offering a hands-on approach to mastering math concepts in a fun and interactive way.

5.References

1. Raspberry Pi Pico.(n.d.). Retrieved from [RaspberryPi]
(<https://www.raspberrypi.org/products/raspberry-pi-pico/>)
2. D82- N52 neodymium magnets
(<https://www.kjmagnetics.com/proddetail.asp?prod=D82-N52>)
3. D21-N52 neodymium Magnets
(<https://www.kjmagnetics.com/proddetail.asp?prod=D21-N52>)